

# Regular Distance-Magic Graphs on Eight Vertices and Generating $\mathbb{F}_2^3$ -Magic Maps

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## Abstract

We classify the finite simple undirected regular ordinary distance-magic graphs on exactly eight vertices, allowing disconnected graphs and degree zero. Up to isomorphism they are  $8K_1$ ,  $2C_4$ ,  $K_{4,4}$ , and  $K_{2,2,2,2}$ . Their reduced adjacency nullities over  $\mathbb{F}_2$  are, in the same order, 7, 4, 6, 4. A direct coordinate argument shows that reduced nullity at least two is equivalent to the existence of a generating  $\mathbb{F}_2^3$ -magic map. Consequently every graph in the classification has such a map. The classification is certified by an exact census of all feasible regular isomorphism classes, all 8! ordinary labelings of every representative, finite-field row-reduction certificates, and a separate labeled-graph coverage calculation.

## 1 Statement and context

For a graph  $G = (V, E)$ , an *ordinary distance-magic labeling* is a bijection  $\ell : V \rightarrow \{1, \dots, |V|\}$  for which

$$\sum_{u \in N_G(v)} \ell(u)$$

is independent of  $v$ . If  $\Gamma$  is a finite abelian group, a  $\Gamma$ -magic map is a map  $f : V \rightarrow \Gamma$  whose open-neighborhood sums are constant in  $\Gamma$ ; it is *generating* when its values generate  $\Gamma$ . A generating map need not be injective.

Batal introduced generating and affinely generating group-magic maps and gave a general reduced-adjacency characterization for regular graphs [1]. He also formulated a generating group distance-magic conjecture and proved its regular case when the labeling group has at most two generators, with a cube-free-order consequence. Those results do not cover the order-eight group  $\mathbb{F}_2^3$ . The result below resolves precisely that regular order-eight layer; it makes no assertion about nonregular graphs, bijective  $\mathbb{F}_2^3$ -labelings, or other orders.

**Theorem 1.** *Let  $G$  be a finite simple undirected regular graph on eight vertices; connectedness and positive degree are not assumed. Then  $G$  is ordinary distance-magic if and only if, up to isomorphism,*

$$G \in \{8K_1, 2C_4, K_{4,4}, K_{2,2,2,2}\}.$$

*Each of these four graphs admits a generating  $(\mathbb{Z}/2\mathbb{Z})^3$ -magic map. Thus every regular ordinary distance-magic graph on eight vertices has such a map.*

The closest inspected prior work is the general algebraic framework just cited. The contribution relative to it is an exact finite classification and certificate at order eight, not a resolution of

the general conjecture. Recent studies of unrelated finite graph and configuration questions also use canonical enumeration, exact arithmetic, and standalone verification [3, 4]; neither supplies a distance-magic result used below. We make no priority claim beyond this bounded comparison.

## 2 The reduced binary criterion

Let  $A$  be the adjacency operator on  $\mathbb{F}_2^V$ , let  $L = \langle \mathbf{1} \rangle$ , and put  $Q_G = \mathbb{F}_2^V / L$ . If  $G$  is regular, then  $A(L) \subseteq L$ , so  $A$  induces

$$\bar{A}_2 : Q_G \longrightarrow Q_G.$$

**Lemma 2.** *A regular graph  $G$  admits a generating  $\mathbb{F}_2^3$ -magic map if and only if*

$$\dim_{\mathbb{F}_2} \ker \bar{A}_2 \geq 2.$$

*Proof.* Put  $S = \{x \in \mathbb{F}_2^V : Ax \in L\}$ . Write a map  $f : V \rightarrow \mathbb{F}_2^3$  by its three coordinate columns  $x_1, x_2, x_3 \in \mathbb{F}_2^V$ . Its neighborhood sums are constant exactly when  $x_i \in S$  for all  $i$ . Its values span  $\mathbb{F}_2^3$  exactly when the matrix with rows  $f(v)$  has rank three, equivalently when  $x_1, x_2, x_3$  are linearly independent. Hence a generating magic map exists exactly when  $\dim S \geq 3$ .

Regularity gives  $L \subseteq S$ , and the quotient map identifies  $S/L$  with  $\ker \bar{A}_2$ . Therefore

$$\dim S = 1 + \dim \ker \bar{A}_2,$$

which proves the equivalence. This is the present elementary-specialization of the general criterion in [1].  $\square$

## 3 Exact classification

Suppose that  $G$  is  $k$ -regular and has ordinary magic constant  $c$ . Summing its eight neighborhood equations counts every label exactly  $k$  times, so

$$8c = k(1 + \cdots + 8) = 36k.$$

Thus  $c = 9k/2$ , whence  $k$  is even. Since  $0 \leq k \leq 7$ , only  $k \in \{0, 2, 4, 6\}$  can occur.

For each feasible degree, the nauty census program **geng** [2] generated all unlabeled representatives. Completeness was then checked without using the generator's class count. For every displayed representative, enumerating all  $8!$  relabelings gives its canonical 28-bit adjacency code and automorphism order. The resulting orbit sizes sum to the number of labeled regular graphs obtained independently by the following exact recursion: process vertices in order, choose the required remaining neighbors of the current vertex among later vertices, decrement their degree demands, and memoize the remaining demand vector. Edges to earlier vertices are already fixed, so every labeled graph is reached exactly once. The two counts agree degree by degree:

| $k$ | representatives | orbit-size sum | recursive labeled count |
|-----|-----------------|----------------|-------------------------|
| 0   | 1               | 1              | 1                       |
| 2   | 3               | 3507           | 3507                    |
| 4   | 6               | 19355          | 19355                   |
| 6   | 1               | 105            | 105                     |

The canonical codes are distinct within each degree. Consequently the eleven representative orbits are pairwise disjoint and exhaust all feasible regular graphs. In the hexadecimal codes below, the unordered pairs  $(i, j)$  with  $0 \leq i < j \leq 7$  are ordered lexicographically, the first pair is the least significant bit, and the minimum over all vertex relabelings is used.

Every one of the  $8! = 40320$  bijections to  $\{1, \dots, 8\}$  was tested on each representative using exact integer neighborhood sums. On the quotient  $Q_G$ , the basis  $[e_0], \dots, [e_6]$  was used; the image of each basis vector was normalized to have last coordinate zero. Thus the stored reduced matrix has entries

$$B_{ij} = A_{ij} + A_{7j} \pmod{2}, \quad 0 \leq i, j \leq 6.$$

Exact row reduction gives the following complete table. “Labels” is the number of ordinary distance-magic bijections, and the last two columns refer to  $\overline{A}_2$ .

| class | canonical hex code | $k$ | labels | rank/nullity |
|-------|--------------------|-----|--------|--------------|
| d0-1  | 0000000            | 0   | 40320  | 0/7          |
| d2-1  | 00cd860            | 2   | 384    | 3/4          |
| d2-2  | 00d5460            | 2   | 0      | 5/2          |
| d2-3  | 04d10e0            | 2   | 0      | 6/1          |
| d4-1  | 03fde78            | 4   | 4608   | 1/6          |
| d4-2  | 077be78            | 4   | 0      | 3/4          |
| d4-3  | 0eb7d78            | 4   | 0      | 5/2          |
| d4-4  | 0ef7c71            | 4   | 0      | 6/1          |
| d4-5  | 2e6b9f8            | 4   | 0      | 3/4          |
| d4-6  | 2e6bb72            | 4   | 0      | 5/2          |
| d6-1  | 7bfdffe            | 6   | 384    | 3/4          |

The four positive representatives have the structures asserted in Theorem 1. Explicit ordinary labelings are easiest to give by their parts. For  $2C_4$ , view each  $C_4$  as  $K_{2,2}$  and use bipartition sides

$$(\{1, 8\} \mid \{3, 6\}), \quad (\{2, 7\} \mid \{4, 5\});$$

each opposite side sums to 9. For  $K_{4,4}$ , label the two parts by  $\{1, 2, 7, 8\}$  and  $\{3, 4, 5, 6\}$ , each of sum 18. For  $K_{2,2,2,2}$ , use independent parts

$$\{1, 8\}, \{2, 7\}, \{3, 6\}, \{4, 5\};$$

every neighborhood then has sum  $36 - 9 = 27$ . Any labeling works for  $8K_1$ , with constant zero.

There are also short structural checks at three degrees. A 2-regular eight-vertex graph is one of  $C_8$ ,  $C_5 \cup C_3$ , or  $2C_4$ . Along any component, the equations  $a_{i-1} + a_{i+1} = 9$  imply  $a_{i+2} = 9 - a_i$  and hence  $a_{i+4} = a_i$ ; bijectivity leaves only  $2C_4$ . Degree zero has one class. The complement of a 6-regular graph is a perfect matching, giving the unique class  $K_{2,2,2,2}$ . The only remaining exclusion is therefore the exact six-class degree-four calculation in the table.

For the four positive classes, their full adjacency ranks over  $\mathbb{F}_2$  are 0, 4, 2, 4. In the last three cases  $\mathbf{1}$  lies in the adjacency image, so passing to the quotient by  $L$  lowers the image dimension by one. The reduced ranks are therefore 0, 3, 1, 3, and the reduced nullities are 7, 4, 6, 4. Lemma 2 now proves the generating-map assertion and completes the proof of Theorem 1.

## 4 Certificates, verification, and scope

The companion data record, for each representative, its adjacency matrix, edge list, ordinary-label count and first witness, reduced matrix, elementary row-operation transcript, reduced row-echelon

form, and a kernel basis. A separate verifier obtains the adjacency matrices through `showg` rather than the producer’s `graph6` decoder, repeats all 443520 labeling tests, recomputes every binary rank with a separate elimination routine, checks the kernel vectors, and replays the row operations. A second pure integer verifier performs the orbit-stabilizer coverage computation and the labeled degree-demand recursion described above. No floating-point arithmetic is used.

The reproducibility bundle contains the `graph6` stream and the scripts `enumerate_order8.py`, `verify_order8_certificate.py`, and `verify_unlabeled_coverage.py`, together with their JSON outputs. The finite computations certify only the order-eight statement. In particular, this article does not settle Batal’s general generating group-distance-magic conjecture, and the generating maps here are not claimed to be affine-generating or bijective.

An AI system assisted with organizing the exhaustive search, implementing certificate and verifier code, and drafting the exposition. The mathematical claims used here are supported by the explicit exact arguments and reproducible integer/ $\mathbb{F}_2$  computations described above; language-model output is not treated as proof. The author retains responsibility for the statement, verification, and manuscript.

## References

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